

Practice Problem Sheet - 6
Transform Calculus
(MA20202)

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Q1. Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 2 \\ 0 & x > 2 \end{cases} \quad \text{Discuss the convergence of the integral at } x=1.$$

Ans: $f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \alpha \cos \alpha (x-1)}{\alpha} d\alpha$; $\int_0^{\infty} \frac{\sin \alpha}{\alpha} d\alpha = \frac{\pi}{2}$

Q2. Find the FIR of the function $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases}$

Ans: $f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin a \alpha \cos \alpha x}{\alpha} d\alpha$

Q3. Represent $f(x) = e^{-kx}$, $x \geq 0$, k a positive constant by
(a) a cosine integral (b) a sine integral.

Ans: FCI of $f(x) = \frac{2k}{\pi} \int_0^{\infty} \frac{\cos \alpha x}{k^2 + \alpha^2} d\alpha$

FSI of $f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\alpha \sin \alpha x}{k^2 + \alpha^2} d\alpha$

Q4. Find the Fourier cosine integral representation for the function

$$f(x) = \begin{cases} x^2 & 0 < x < a \\ 0 & x > a \end{cases}$$

Ans: $f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{1}{\alpha^3} [(a^2 \alpha^2 - 2) \sin \alpha x + 2a \alpha \cos \alpha x] d\alpha$

Q5. Using Fourier integral representation, show that

$$\int_0^{\infty} \frac{\cos \frac{\pi \alpha}{2} \cos \alpha x}{1 - \alpha^2} d\alpha = \begin{cases} \frac{\pi}{2} \cos x & |x| \leq \frac{\pi}{2} \\ 0 & |x| > \frac{\pi}{2} \end{cases}$$

[Hint: Consider $f(x) = \begin{cases} \frac{\pi}{2} \cos x & |x| \leq \frac{\pi}{2} \\ 0 & |x| > \frac{\pi}{2} \end{cases}$ and FCI]

***** The End *****